

TABLE II
Technological Application

Technologies	Description	Short term (2010)	Mid term (2015)	Long term (2020 +)
Materials	Robotic parts and systems are composed or can be made of a variety of materials. As materials R&D is mostly driven by other domains, technology transfer to robotics will be greatly beneficial, particularly in composites, light metal foams and materials integrating functionality such as sensing and actuation.	Shape memory alloys (SMA) and electro-active polymers (EAP) for microrobots; some use of carbon /composite / metal foams	SMA and EAP for robot reconfiguration; biomimetic /sensing materials; some use of nanomaterials	Increased use of nanomaterials; use of biomimetic materials and biological tissue; intelligent materials and structures
Navigation	Navigation is concerned with controlling movement. It relies on mapping, localisation and collision avoidance. Unlike map-based navigation, combining localisation and mapping (SLAM) and collision avoidance are robotics driven.	Navigation expensive (computation and sensors); localisation and mapping in controlled environments solved	Some perception-based localisation; SLAM for challenging environments; collision avoidance considers dynamic objects	SLAM in unconstrained environments; collision avoidance with dynamic, non-cooperative obstacles through perception
Planning	Planning is the computation and selection of paths, motions, actions, tasks, policies, procedures and missions for goal-directed robot behaviour. Most aspects of planning are driven by several industries, each concentrating on their context.	Manual programming superior to automated planning (optimised process path based on human experience); randomised motions as planning alternative	Automated mission and process planning using, for example, databases of expert knowledge	Autonomous, online planning for tasks of high dimensionality; learn from human (often interactively)
Power management	Power management efficiently generates, stores and conditions power for the system. With the exception of power management for sensors, this technology is currently driven by worldwide 'e-mobility' initiatives.	Mostly external power or local storage; regenerative brakes available, but not used often	Local energy conversion/generation; regeneration is standard; planners conserve energy	Efficient wireless power transfer; system efficiency continues to increase
Control	Control uses algorithms and mathematics to regulate the behaviour of devices or systems. Robotics drives the application of control theory developed in other domains to robotics (eg, kinematics, dynamics and force control).	Control through cascades; state-space controller; sliding mode controller; feedback linearisation	Predictive, distributed, self calibrating, self tuning controllers	Fault tolerant controllers; automatic reconfiguration of controllers
Learning	Learning refers to adaptation of robot behaviour through practice, experience or teaching. Basic research on machine learning is often evaluated by robotics, but the web technology, games industries and the AI community are also prime users.	Parts of robot systems use learning methods; well-defined conditions; learning from an expert teacher	Essential parts of controllers use learning methods; learning by experience; learning by demonstration	Complete robotic systems use learning methods (learning by observation, flexible conditions)
Modelling	Modelling is the mathematically described approximation of reality. Most of the modelling is driven by other domains, but robotics has a strong need to model and simulate the system (mechanics, actuators, electronics and sensors) and environment at runtime.	Lack of standards for model descriptions; simulation not as good as real-world experiments; long computation times	Standard language for model description; interchangeable models; modelling of flexible and soft bodies; improved cybernetics	Real-time, dynamic modelling and interpretation allow for accurate assessment of the robot's and the world's state
Sensors	A sensor detects or measures a physical quantity and converts it into electrical signals. The development of a few sensors (eg, skin sensors) and some sensor properties (eg, size, weight and safety category) are robotics driven.	Gradual replacement of special hardware (frame grabbers, cameras...); 3D vision sensors in low resolution	Higher frame rate of visual sensors; greatly improved 3D vision sensors; no moving parts in laser scanners	Advanced task-dependent sensor fusion; multiple sensor modalities; step change in visual servoing; known event interpreted
Sensing and perception	Perception is the robot's ability to build and interpret representations of the physical world from sensed data. This process may involve cognition and learning. Sensing is not robotics driven, but perception under real-time constraints and fusing often uncertain information from many sources are.	Sensor fusion is task-specific and relies on calibration; limited by processing power; use of attention mechanisms	Visual processes on sensor or dedicated processors; multimodal sensing for intrinsic safety	Sensing on chip; perception techniques take over from fusion (closer to human perception system); no longer task dependent